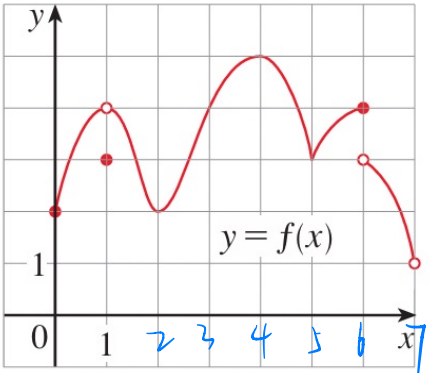
Exercises 4.1

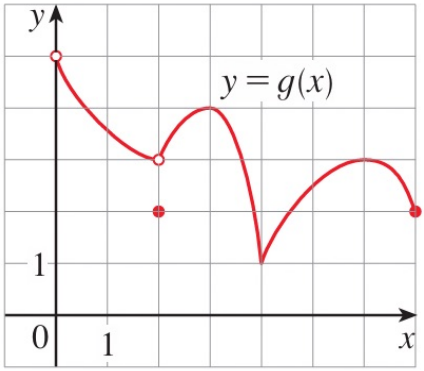
5, 17, 27, 47, 53, 57.

5–6 Use the graph to state the absolute and local maximum and minimum values of the function.

5.



6.



not defined

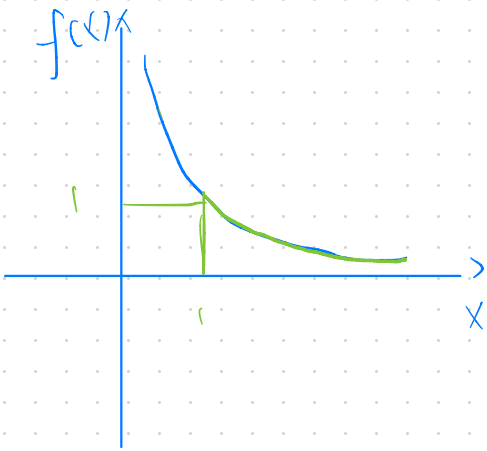
$f(7)$ is abs min, no abs min

$f(4)$ is abs max

$f(1)$, $f(2)$, $f(5)$ are local minimum

$f(4)$, $f(6)$ are local maximum

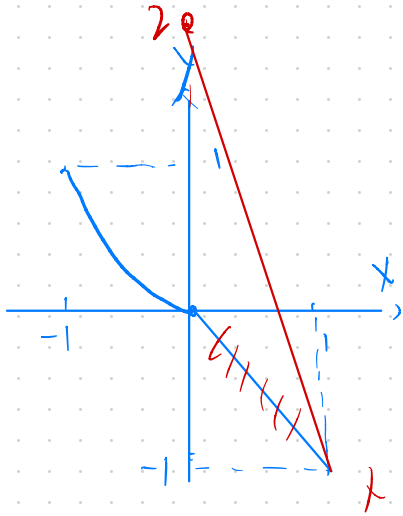
17. $f(x) = \frac{1}{x}$, $x \geq 1$



$f(1) = 1$ is absmax

The is no absmin and local extreme.

27. $f(x) = \begin{cases} x^2, & \text{if } -1 \leq x \leq 0 \\ 2-3x, & \text{if } 0 < x \leq 1 \end{cases}$



no absmax

$f(-1) = 1$ is absmax

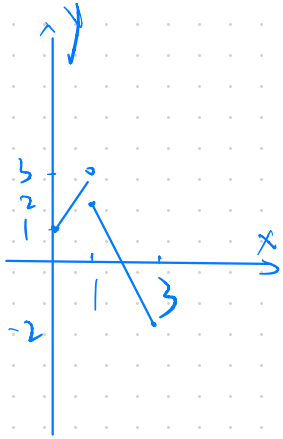
$f(1) = 1$ is absmin

There is no local extreme

$f(0) = 0$ is local minimum

no local maximum.

28 $f(x) = \begin{cases} 2x+1, & 0 \le x < 1 \\ 4-2x, & 1 \le x \le 3 \end{cases}$



abs min: $f(3) = -2$

local minimum: $f(1) = 2$

no local minimum ?

no abs max, local max

47. $g(x) = x^2 \ln x$ Find the critical points

domain $x \in (0, +\infty)$

end points: $f($

$$g'(x) = 2x \ln x + x$$

① stationary point:

$$g'(x) = 2x \ln x + x = 0$$

$$x = e^{-\frac{1}{2}}$$

② No singular point. $g'(x)$ DNE, but 0 is not an interior point of the domain of f.

③ $x = 0$ is not in the domain, no start point.

53. $f(x) = 2x^3 - 3x^2 - 12x + 1$, $[-2, 3]$

end point : $f(-2) = 2(-2)^3 - 3(-2)^2 - 12(-2) + 1$

$= -3$

$f(3) = 2 \times 3^3 - 3 \times 3^2 - 12 \times 3 + 1$

$= -8$

stationary point : $f'(x) = 6x^2 - 6x - 12$

$f'(x) = 0$

$6(x-2)(x+1) = 0$

$x = 2$ or $x = -1$

$f(2) = 2 \times 2^3 - 3 \times 2^2 - 12 \times 2 + 1 = -19$ (absmin)

$f(-1) = 2 \times (-1)^3 - 3 \times (-1)^2 + 12 + 1 = 8$ (absmax)

Singular point : No as $f'(x)$ is defined everywhere.

57. $f(x) = x + \frac{1}{x}$ $[0.2, 4]$

end point: $f(0.2) = 0.2 + \frac{1}{0.2} = 5.2$ (abs max)

$f(4) = 4 + \frac{1}{4} = 4.25$

stationary point: $f'(x) = 1 - x^{-2}$

$f'(x) = 1 - x^{-2} = 0$

$x^{-2} = 1$

$x = 1$

$f(1) = 1 + \frac{1}{1} = 2$ (abs min)

Singular point: No $f'(x)$ DNE when $x = 0$, but 0 is not in the given interval